HW2

Q1 Referred: OR-notes
[people.brunel.ac.uk/~mastjjb/jeb/~~or~~ or/contents.html]

For this problem of defining the LP for optimal assignment of workers to make widgets we have to define —

1. variables
2. constraints
3. objective

1. Variables:-

Each worker can make any widget
The variables ~~belong~~ to are continuous ~~and~~ Hence one worker can make a fraction of widget and another worker can finish the remaining

we have 4 workers and 3 widgets
Let the workers be $(x_1, x_2, x_3, x_4) \in R$
and $(a, b, c) \in R$ be widgets

Notation: $x_{1a} \rightarrow 1a \rightarrow$ worker 1, widget a.

$x_{1a}, x_{1b}, x_{1c} \rightarrow$ widgets made by x_1

$x_{2a}, x_{2b}, x_{2c} \rightarrow$ widgets made by x_2

$x_{3a}, x_{3b}, x_{3c} \rightarrow$ widget made by x_3

$x_{4a}, x_{4b}, x_{4c} \rightarrow$ widget made by x_4

Q1

Each worker works max 35 hrs/week

$35 \times 60 = 2100$ min/week

Constraints for

Time spent by each worker can be expressed as: — (1)

(1) worker Time constraint $Ax \leq b$

worker 1 $5x_{1a} + 6x_{1b} + 13x_{1c} \leq 2100$ — (1)

worker 2: $7x_{2a} + 12x_{2b} + 14x_{2c} \leq 2100$ — (2)

$Ax \leq b$ $4x_{3a} + 8x_{3b} + 9x_{3c} \leq 2100$ — (3)

$10x_{4a} + 15x_{4b} + 17x_{4c} \leq 2100$

The constraint for min widget to be made:- $Wx \geq t$ — (2)

$x_{1a} + x_{2a} + x_{3a} + x_{4a} \geq 100$

$x_{1b} + x_{2b} + x_{3b} + 9x_{3c} \geq 150$

$13x_{1c} + 14x_{2c} + 9x_{3c} + 17x_{4c} \geq 100$

(2) The objective function is minimizing cost:-

$Wx \geq t$

t: Target widget

$x_{1a} + x_{2a} + x_{3a} + x_{4a} \geq 100$

$x_{1b} + x_{2b} + x_{3b} + x_{4b} \geq 150$

$x_{1c} + x_{2c} + x_{3c} + x_{4c} \geq 100$

Q1

The objective function for minimizing the cost of widget Production :-

The cost of production has to be minimized ~~[illegible]~~.

Cost of widget Production for x_1 :-

cost of $x_{1a} = \left(\frac{5 \cdot x_{1a}}{60}\right) 10$

$x_{1b} = \left(\frac{6 \cdot x_{1b}}{60}\right) \cdot 18$

$x_{1c} = \left(\frac{13 \cdot x_{1c}}{60}\right) \cdot 15$

and so on

~~Cost~~

worker (i →) worker i

	1	2	3	4
widget A	(5.10)/60	(7.8)/60	4.6/60	(10) 9/60
j↓ B	(6.18)/60	(12.20)/60	8.15/60	(15) 17/60
C	(13.15)/60	14.16/60	9.13/60	(17) 17/60

(3)

The above is the cost matrix Per widget Per worker → let C_{ij}, $i(1,2,3,4)$, $j(a,b,c)$

The cost objective can be expressed as (3)a

$$\min \sum_{i=1}^{4} \sum_{j=a}^{c} (C_{ij} \cdot x_{ij})$$

Hence it can be written in std form as

∴ Objective → minimize cost

$$\min C_{cost}^{T} X \quad \text{(cost function)}$$

$C^T \in C_{ij}$: cost vector per widget per worker

$X \in X_{ij}$: # of widgets ~~per made~~ made by worker i j

Per contrains imposed by

$$Ax \leq b \quad \text{① Reference earlier}$$

$$Wx \geq t \quad \text{② Reference earlier}$$

all $x_{ij} \geq 0, \in R$ $x \geq \emptyset$

$c_{ij} \geq 0, \in R$ $w_{ij} \geq 0, \in R$ ~~b is the worker~~

$Ax \leq b$ is the work time constraint

$Wx \geq t$ is the widget target constraint

$x \geq \emptyset$.

→ A, and W are sparse matrices

→ → C contain all the element from cost matrix ③ as a column.

→ $X = [x_{1a}, x_{1b}, x_{1c}, x_{2a}, x_{2b} \cdots x_{4c}]^T$

Q1

$X = [x_{1a}, x_{1b}, x_{1c}, x_{2a}, \dots x_{4c}]^T$

A is a matrix of the form (sparse matrix)

$$A = \begin{bmatrix} 5 & 6 & 13 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 7 & 12 & 14 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 8 & 9 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10 & 15 & 17 \end{bmatrix}$$

Similarly W

$$W = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

As stated above $\min C^T X$

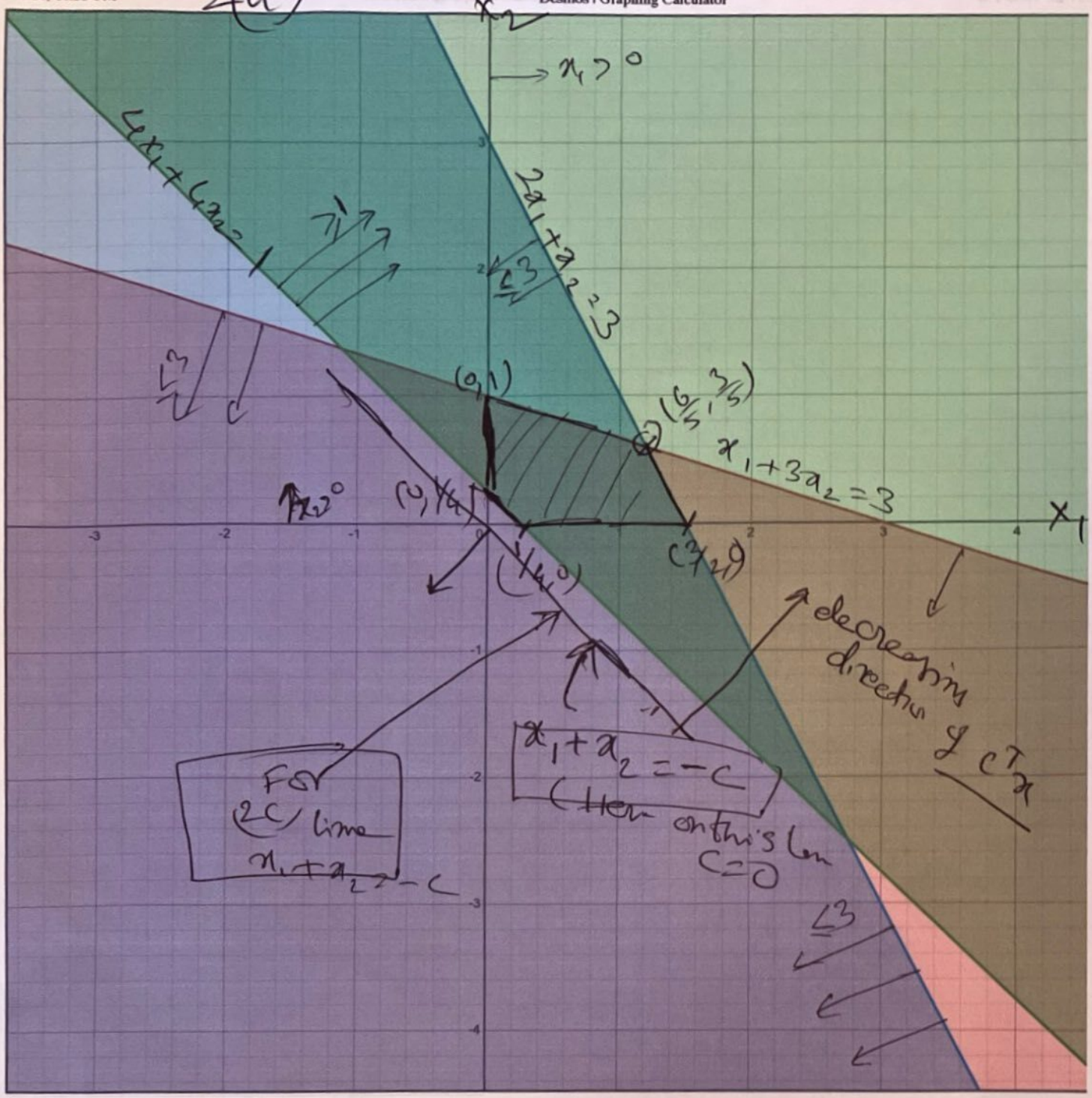
s.t $Ax \leq b$ (Time)

$Wx \geq t$ (Production target)

$X \geq 0$

$t \geq 0$

2(a)



1 ~~$x+3y \leq 3$~~ $x_1 + 3x_2 \leq 3$

2 ~~$2x+y \leq 3$~~ $2x_1 + x_2 \leq 3$

3 $4x + 4y \geq 1$ $4x_1 + 4x_2 \geq 1$

4 ~~$x \geq 0, y \geq 0$~~ $x_1, x_2 \geq 0$

5

(2a)

2(c)

2b)

$$\min : -x_1 - x_2$$

$$\text{s.t.} \quad x_1 + 3x_2 \le 3$$
$$2x_1 + x_2 \le 3$$
$$4x_1 + 4x_2 \ge 1$$
$$x_1, x_2 \ge 0$$

$$x_1 + 3x_2 = 3$$
$$\cancel{2x_1 + x_2 = 3}$$
$$6x_1 + 3x_2 = 9$$
$$+5x_1 = +6$$
$$x_2 = 1.2, \; x_2 =$$
$$x_1 = \frac{6}{5}, \; x_2 = \frac{3}{5}$$

$$\boxed{\left(\frac{6}{5}, \frac{3}{5}\right)}$$ 1

$$x_1 + 3x_2 = 3$$
$$\cancel{4x_1 + 4x_2 = 1}$$
$$x_1 + x_2 = 1/4$$
$$2x_2 = \frac{11}{4}$$
$$x_2 = \frac{11}{8}$$
$$\cancel{x_1 = 3 - 3}$$
$$x_1 = \frac{1}{4} - \frac{11}{8} = -\left(\frac{9}{8}\right)$$

$$\boxed{(x_1, x_2) = \left(-\frac{9}{8}, \frac{11}{8}\right) \quad \text{Not feasible } x_1 < 0}$$

$$2x_1 + x_2 = 3$$
$$\cancel{4x_1 + 4x_2 = 1}$$
$$x_1 + x_2 = 1/4$$

$$\boxed{x_1 = \frac{11}{4}, \; x_2 = -\frac{10}{4} \quad \text{Not feasible}}$$

$$x_1 + 3x_2 = 3$$
$$x_1 = 0, \; x_2 = 1$$
$$x_1 = 3, \; x_2 = 0$$

2 ~~extreme~~ Points

$$2x_1 + x_2 = 3$$
$$x_1 = 0, \; x_2 = 3$$
$$\boxed{x_1 = \frac{3}{2} \quad x_2 = 0}$$ 3

$$4x_1 + 4x_2 = 1$$
$$\Rightarrow x_1 = 0 \; x_2 = 1/4$$
$$x_1 = 1/4 \; x_2 = 0$$

4

5

2b

Extreme Points:-	$(-x_1 - x_2)$
$(\frac{6}{5}, \frac{3}{5})$	$-\frac{9}{5}$
$(0, 1)$	-1
$(0, 1/4)$	$-\frac{1}{4}$
$(1/4, 0)$	$-\frac{1}{4}$
$(3/2, 0)$	$-3/2$

optimal value for <u>min</u> $(-x_1 - x_2)$

at $(\frac{6}{5}, \frac{3}{5})$ - Has an optimal

$\boxed{\text{optimal value of } -\frac{9}{5}.}$

2c

$$\min(-x_1 - x_2)$$

In level set approach

$$-x_1 - x_2 = c$$

$$\Rightarrow x_1 + x_2 = -c$$

2c)

$\cancel{x_1 + x_2 = c}$

$x_1 + x_2 = -c$

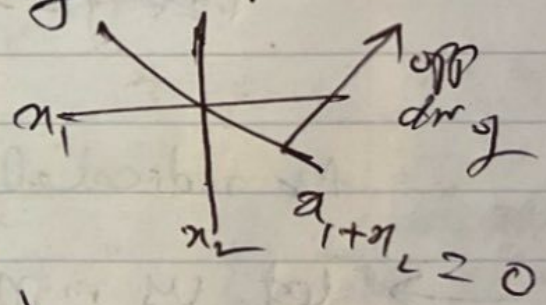
$x_1 \geq 0$ and $x_2 \geq 0$

as x_1 and x_2 increase $\Rightarrow$ c decrease

i.e. c minimizer.

The line $x_1 + x_2 = -c$ is a line with a slope of -1

(see attached graph)



$\cancel{\min(x}$ $\min(-x_1, -x_2)$

$\nabla(\min(-x_1, -x_2)) = (-1, -1)$

As we move in the opp. direction of gradient toward $(1,1)$, $(2,2)$ ... the value of $c^T x$ decrease

The last time we encounter feasible set is at $(\frac{6}{5}, \frac{3}{5})$ Hence $-\frac{9}{5}$ is optimal.

3a Formulate following as LP:-

$\min \|Ax - b\|_1$ subj to $\|x\|_\infty \leq 1$

$A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $x \in \mathbb{R}^n$

While L_1 and L_∞ norms are convex, the norms L_1 norm, and L_∞ norm are not linear

As in Lecture 3.4 (ABS value 1/2)

Let us introduce z, and state to convert to linear

our problem as $\min : z$, $z \in \mathbb{R}^m$

$$z \geq \|Ax - b\|_1$$

implies $z_i \geq (Ax - b)_i$ and $z_i \geq -(Ax - b)_i \; \forall i$ ①

~~Can be written as Lets say~~ $\Rightarrow$ $(i \to 1 \text{ to } m)$. ~~$(i \to 1 \text{ to } m)$~~

subjected to

$\Rightarrow \|x\|_\infty \leq +1$ can be stated as (a sating norm)

$$-1 \leq x_j \leq 1, \quad \forall j - 1 \text{ to } n$$

$\boxed{L1 \text{ norm } \|x\|_1 = \sum_{i=1}^{n} |x_i| \text{ if } x \in \mathbb{R}^n}$ $\Rightarrow$ $\min \sum_{i=1}^{m} z_i$ $(i = 1 \text{ to } m)$

s.t. $(Ax - b)_i \leq z_i$, $i : 1 \text{ to } m$

$-(Ax - b)_i \leq z_i$

$z_i \geq 0$

$-1 \leq x_j \leq 1$ $(j = 1 \text{ to } n)$

3.5 Formulate LP

$\min \|x\|_1$ subj to $\|Ax-b\|_\infty \le 1$

$A \in R^{m \times n}$, $b \in R^m$, $x \in R^n$

Same principle as 3(a) L_1 and L_∞ norms are convex but not linear

As in lecture 3.4, to convert to linear (let's introduce z,

As in lecture 3.4 Abs value e.g. $1/2$)

our problem is minimize z, $z \in R^m$

so that $z \geq \|x\|_1$

implies $z_i \geq x_i$ and $z_i \geq -x_i \ \forall i$

$\left(i = 1 \text{ to } n\right)$ i $\left.\begin{array}{l} x_i \le z_i \\ -x_i \le z_i \end{array}\right\}$

subjected to

$\|Ax-b\|_\infty \le 1$ gated by infinity norm

$-1 \le (Ax-b)_j \le 1 \ \forall j \ (1 \text{ to } m)$

The LP formulation can be stated as

$$\left[\begin{array}{l} \min \sum_{i=1}^{n} z_i \\ \quad \text{st. } \left.\begin{array}{l} x_i \le z_i \\ -x_i \le z_i \end{array}\right\} i = 1 \text{ to } n \\ \quad -1 \le (Ax-b)_j \le 1, \ j = 1 \text{ to } m \\ \quad z_i \ge 0 \end{array}\right.$$

3C Formulate LP

$$\min \|Ax-b\|_1 + \|x\|_\infty$$

Characterizing $\|Ax-b\|_1$ and $\|x\|_\infty$ are expanded in 3(a)

~~$\|x\|_\infty$ is~~

$\|Ax-b\|_1$ is expanded in 3(a)

Let us introduce variable z so that

$$z \geq \|Ax-b\|_1 \text{ as in } 3(a)$$

unlike 3(a) where infinity norm ~~is bounded~~, is bounded in $[-1,1]$ Here

$$\|x\|_\infty = \text{maximum } |x_i|$$

As in lecture 3.4, let us introduce variable y,

$$y \geq \|x\|_\infty \Rightarrow y \geq |x_i|$$

$$\Rightarrow y \geq x_j \text{ and } y \geq -x_j$$

$$\Rightarrow x_j \leq y \text{ and } -x_j \leq y$$

$\forall j, j: 1 \leq n$.

The LP formulation

$$\min \sum_{i=1}^{m} z_i + y$$

$$\text{s.t } (Ax-b)_i \le z_i, \quad i: 1 \text{ to } m$$

$$-(Ax-b)_i \le z_i, \quad i: 1 \text{ to } m$$

$$z_i \ge 0$$

$$x_j \le y, \quad j: 1 \text{ to } n$$

$$-x_j \le y, \quad j: 1 \text{ to } n$$

(4) Formulate LP :-

$A \in R^{m \times n}$, $b \in R^m$

$$\min_{x \in R^n} \sum_{i=1}^{m} \max\{0, a_i^T x + b_i\}$$

Same principle as lecture 3.4, e.g. Abs 1/2

$\sum_{i=1}^{m} \{0, a_i^T x + b_i\}$ is convex but not linear. (like Relu)

To convert to linear, so that it can be minimized, let's introduce variable z, $z \in R^m$ so that

$$z_i \geq 0$$
$$z_i \geq a_i^T x + b_i \quad \forall x, i: 1 \leq m$$

& $0 \leq z_i$ and $a_i^T x + b_i \leq z_i$ $\forall x$, $i: 1 \leq m$

$0 \leq z_i$ always kicks in when $a_i^T x + b_i \leq 0$

LP can be formulated as:

$$\min \sum_{i=1}^{m} z_i$$

S.T ~~$0 \leq z_i$, $i: 1 \leq m$~~

$a_i^T x + b_i \leq z_i$, $i: 1 \leq m$

$z_i \geq 0$ $i: 1 \leq m$, $x \in R^n$, $z \in R^m$

6a $\min c^T x$, s.t $0 \le x \le 1$, $x \in \mathbb{R}^n$, $c \in \mathbb{R}^n$

$x \in [0, 1]$, $x \in \mathbb{R}^n$

$c^T x = c_1 x_1 + c_2 x_2 + c_3 x_3 + \dots + c_n x_n$

~~This is~~

$c^T x$ is min with sum as follows

$x_i = +1$ for all $c_i < 0$

$x_i = 0$ for all $c_i > 0$

$x_i \in [0, 1]$ for $c_i = 0$

This can be written as

optimal value $\min c^T x = \left\{ \sum_{i=1}^{n} c_i x_i \right.$ $\begin{cases} x_i = 1, \text{ if } c_i < 0 \\ x_i = 0, \text{ if } c_i > 0 \\ x_i \in [0,1], \text{ if } c_i = 0 \end{cases}$

optimum soln

$\min c^T x$ is unique (assuming all $c_i \neq 0$)

6(b) $\min c^T x$, s.t. $-1 \le x \le 1$, $x \in \mathbb{R}^n$, $c \in \mathbb{R}^n$

$x \in [-1, 1]$

Just as in 6(a) the optim value and solution can be written ~~are~~ similarly

Given $x \in [-1, 1]$, $\Rightarrow x_i = -1$ for $c_i > 0$ and $x_i = 1$ for $c_i < 0$ to get min

$\min c^T x = \left\{ \sum_{i=1}^{n} c_i x_i \right.$ $\begin{cases} x_i = 1 \text{ if } c_i < 0 \\ x_i = -1 \text{ if } c_i > 0 \\ x_i \in [-1, 1], \text{ if } c_i = 0 \end{cases}$

optim. value

The min is unique (assuming all $c_i \neq 0$.)

$$\min c^T x, \text{ s.T } -1 \leq 1^T x \leq 1, \cancel{x \in R^n, c \in}$$
$$x \in R^n, c \in R^n$$

6(C) $1^T x = \sum_{i=1 \text{ to } n} x_i \rightarrow$ sum of all elements of x

Sum of all elements $\sum_{i=1 \text{ to } n} x_i \in [-1, 1]$

However any combination of x_i in x can lead to a sum of $+1$, or -1

$+1$ or -1 is stated because

$$\min c^T x = \min \sum_{i=1}^{n} c_i x_i \text{ subject}$$
$$\text{to } \sum x_i \in [-1, 1]$$

~~any set~~ or For e.g. if we take R^n, $n = 2$, in R^2

any vectors such as $[19, -20]$ or $[100, -100]$ or $[-1, 0]$ all the vectors of x will work.

$[-1, +1]$ for $[19, -20]$

$[\cancel{\ }, +1]$ for

$c = [1, 1]$ for $x = [19, -20]$

$c = [1, 1]$ for $x = [100, -100]$

6(C)

The solution set is $\boxed{\text{Infinite in } R^2}$
similarly in R^n the solution set is
Infinite in R^n

$\min c^T x \quad s.t. -1 \leq 1^T x \leq 1$

feasible set $\sum_{i=1 \text{ to } n} x_i \in [-1, 1]$

In generic form we can state as

If all $C_i \geq 0$
optimal solution $x_i = \begin{cases} -1 & \text{on } \max(C_i) \\ 0 & \text{all other} \end{cases}$

optimal value $\min c^T x = -\max(C_i)$

If all $C_i \leq 0$
optimal solution $x_i = \begin{cases} +1 & \text{on } \min(C_i) \\ 0 & \text{all oth} \end{cases}$

optimal value $\min c^T x = \min(C_i)$

If all $C_i = 0$ optimal solution $x_i = \begin{cases} \text{any distro} \\ \text{to fit} \end{cases}$
$-1 \leq 1^T x \leq 1$

optimal value $\min c^T x = \phi$

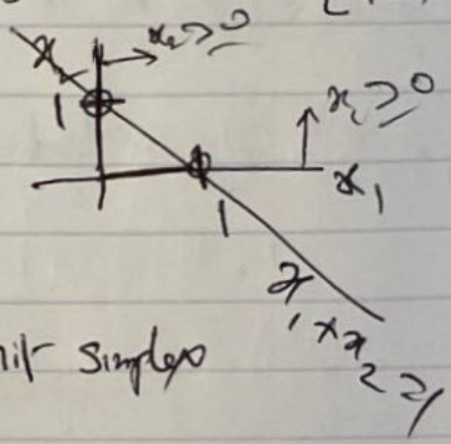
mixed sign C_i, leads to unbounded scenario

6d $\quad \underset{x \in \mathbb{R}^n}{\text{minimize}}\ c^T x \quad \text{s.t.}\ 1^T x = 1,\ x \geq 0$

(d) $1^T x = \sum_{i=1}^{n} x_i$

In $\mathbb{R}^2$, $1^T x \Rightarrow$ geometrically

$$\Rightarrow 1^T x = x_1 (1^T e_1) + x_2 (1^T e_2)$$
$$= x_1 \left(\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) + x_2 \left(\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$
$$= x_1 + x_2$$



In 3-D this a Δ^{lar} plane cutting, x, y, z axes

$x \geq 0$, $1^T x$ <u>forms</u> a unit simplex

The vertices are of the form

In 2D – $(0, 1)$, $(1, 0)$

3D – $(0\ 0\ 1)$ $(0\ 1\ 0)$ $(1\ 0\ 0)$

Feasible set constitutes the vertices

optimum value

$$\therefore \min c^T x = \min_{i=1 \text{ to } n} (c_i)$$

corresponding <u>optimal</u> solution $x_i = \begin{cases} 1 & i \in \min c_i \\ 0 & \text{for all the other } c_i \end{cases}$

(*) The solution need not <u>be unique</u>

6e

$$\text{maximize } c^T x,$$
$$\text{s.t. } 1^T x = k, \quad 0 \le x \le 1$$
$$x \in \mathbb{R}^n, \quad k \text{ is integer}, \quad 1 \le k \le n$$

$x \in \mathbb{R}^n$, $1 \le k \le n$.

$1^T x = k$ indicates that, similar to $1^T x = 1$ as in 6(d), ~~this is a~~

$x_1 + x_2 + \cdots x_n = k$ is a Hyperplane

$0 \le x \le 1$, x lies between 0 and 1.

If $x_1, x_2, \cdots x_k = 1$

$x_{k+1}, x_{k+2}, \ldots x_n = 0$

Putting the top k-values of c_i on x_1 to x_k maximizes $c^T x$

This can be written as

$$\max c^T x = \begin{cases} x_i = 1, \rightarrow & \text{For every } c_i \text{ that is among TOP-K max values of } c_i \\ x_i = 0 & \text{For every } c_i \text{ that is not among TOP-K max values of } c_i \end{cases}$$

The optimal solution need not be unique.